MGRK UNIVERSITY

Cloud Monitoring & Load Analysis Report

Project Title

Proactive ML-Based Load Balancing for Cloud Microservices Using LSTM, Reinforcement Learning and Kubernetes.

About MGRK University

MGRK University provides a modern, technology-driven academic environment that supports digital education and cloud-based student services. The Student Result Management System allows students to securely access their academic results, branch rankings, and analytics through a centralized cloud platform. This report presents the monitoring and load analysis of the system during the latest load-monitoring session.

Project	MGRK University Student Result Portal
Report Date	2026-10-06
Generated Date	06-10-2026
Generated Time	18:58:53
Monitoring Duration	0 days 00:32:23
Data Points	315
Generated By	Cloud Monitoring System
Version	2.0

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System Summary

Highest CPU	100.00%
Average CPU	72.22%
Highest Memory	100.00%
Average Memory	78.23%
Highest Requests/sec	120.46
Average Requests/sec	87.22
Highest Response Time	6.011 sec
Average Response Time	0.050 sec
Peak Users	500

Peak Load Analysis

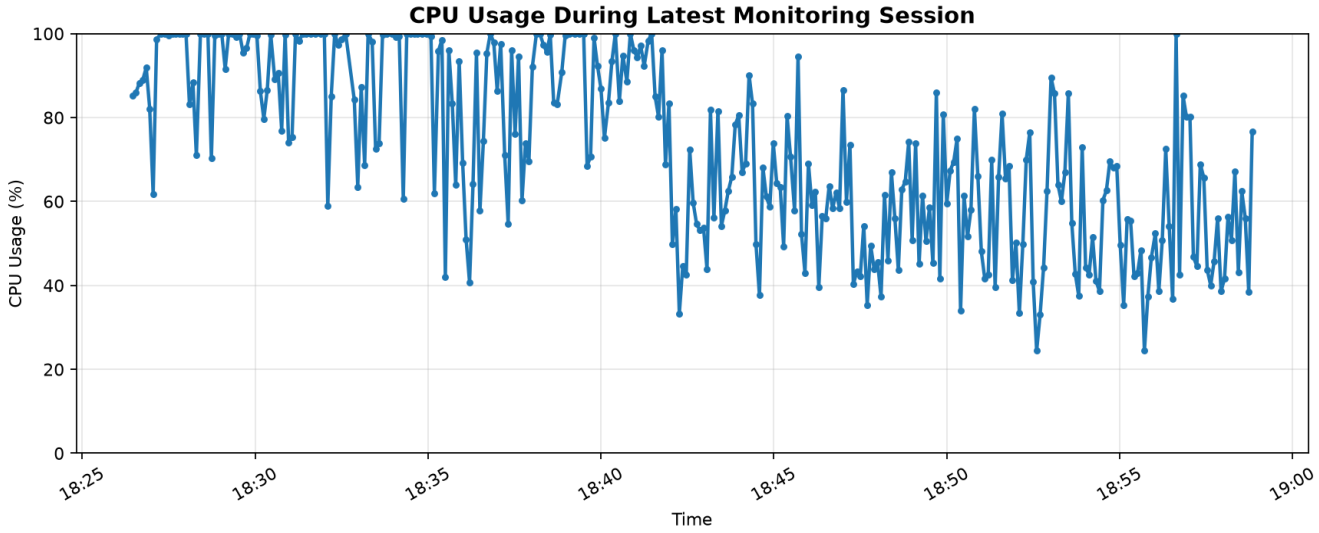
Peak Time	2026-10-06 18:27:16
CPU	100.00%
Memory	81.90%
Requests/sec	12.87
Response Time	0.078 sec
Current Users	400
Load Level	Very High
Most Requested API	/result

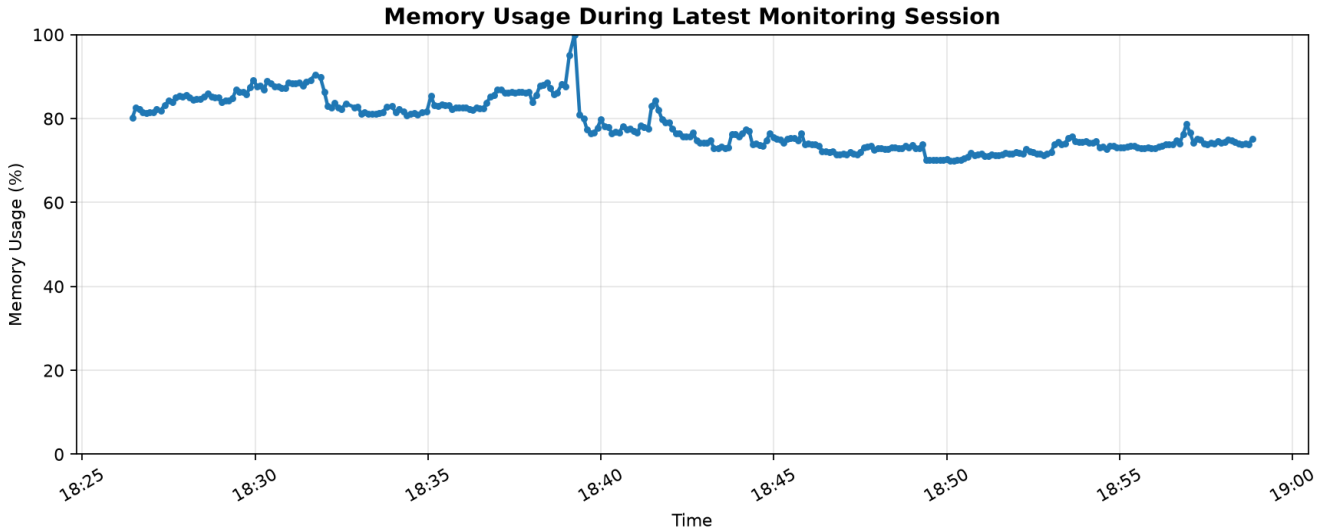
API Usage Analysis

API	Samples	Percentage
/result	218	69.21%
/branch-ranks	59	18.73%
/college-ranks	38	12.06%

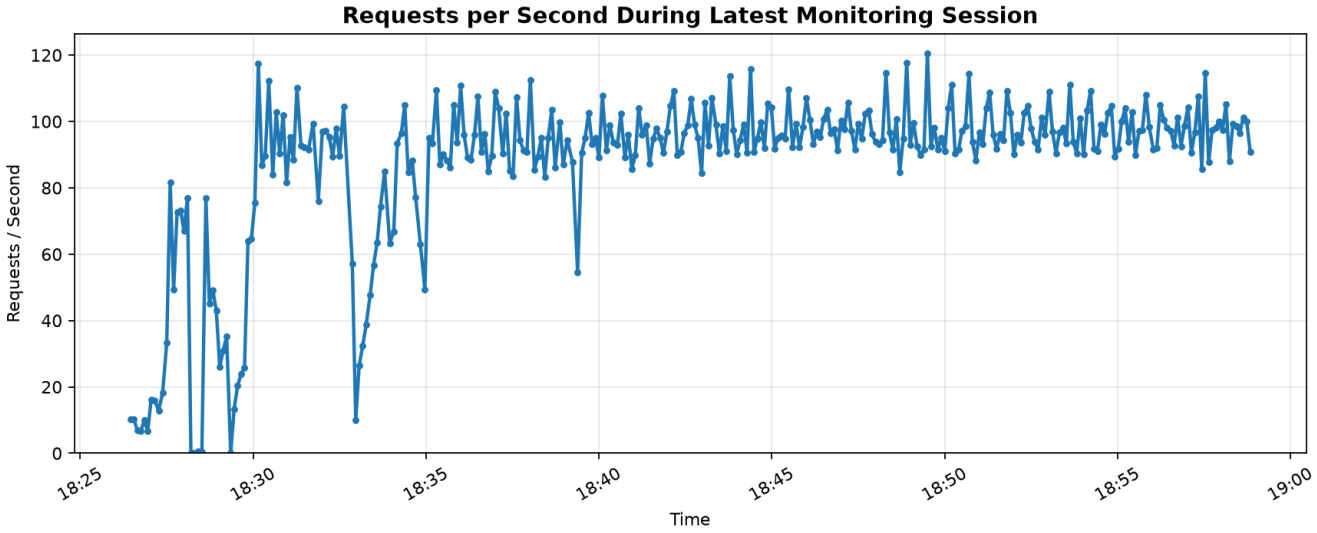
API usage percentages represent the distribution of monitored traffic samples by the API recorded as the most requested endpoint at each collection interval.

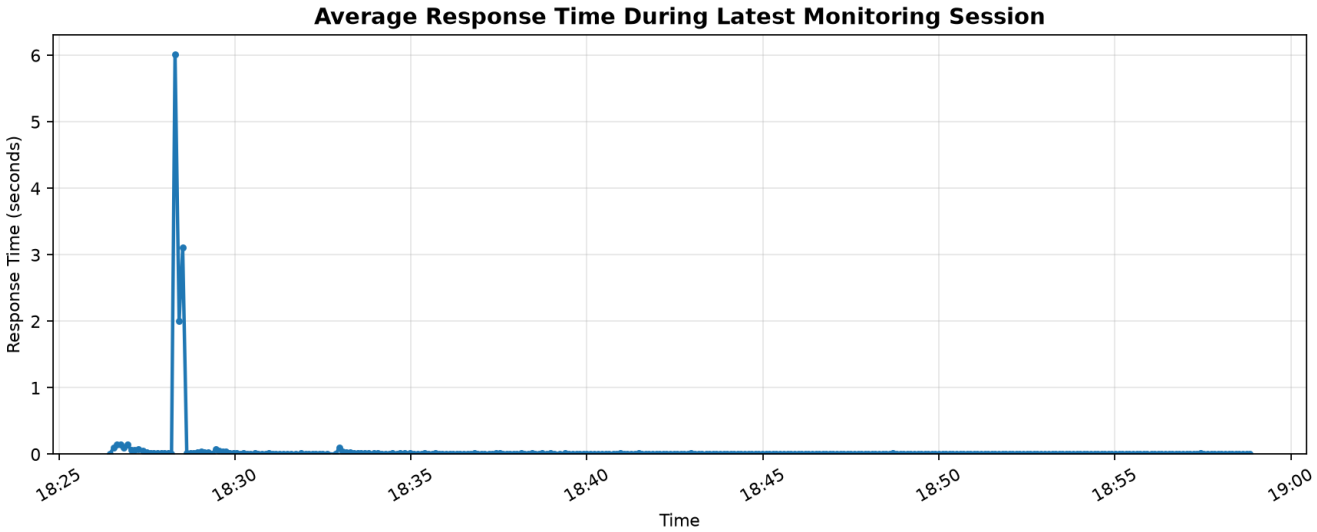
Performance Graphs





Traffic Analysis





Overall Analysis & Recommendations

Monitoring Period
2026-10-06

Overall Analysis

Highest CPU Usage : **100.00%**

Average CPU Usage : **72.22%**

Highest Memory Usage : **100.00%**

Average Memory Usage : **78.23%**

Peak Users : **500**

Highest Requests/sec : **120.46**

Average Requests/sec : **87.22**

Highest Response Time : **6.011 seconds**

Average Response Time : **0.050 seconds**

Recommendations

• Monitor CPU and Memory continuously.

• Predict traffic before peak load using LSTM.

• Scale Kubernetes Pods automatically during High and Critical load.

• Use the predicted CPU metric with Kubernetes autoscaling to support proactive scaling.

• Continue collecting time-series data to improve prediction accuracy.

• Compare predicted CPU with actual CPU during future monitoring sessions.

End of Report